

# Mathematics for Information Technology

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## Integration by Parts

### • OBJECTIVES

- Find an antiderivative using **integration by parts**.

## Integration by Parts

This technique can be applied to a wide variety of functions and is particularly useful for integrands involving products of algebraic and transcendental functions. For instance, integration by parts works well with integrals such as

$$\int x \ln x \, dx, \quad \int x^2 e^x \, dx, \quad \text{and} \quad \int e^x \sin x \, dx.$$

Integration by parts is based on the formula for the derivative of a product

$$\frac{d}{dx}[uv] = u \frac{dv}{dx} + v \frac{du}{dx} = uv' + vu'$$

where both  $u$  and  $v$  are differentiable functions of  $x$ . If  $u'$  and  $v'$  are continuous, you can integrate both sides of this equation to obtain

$$uv = \int uv' \, dx + \int vu' \, dx = \int u \, dv + \int v \, du.$$

$$\begin{aligned} \frac{dv}{dx} = v' &\rightarrow dv = v' dx \\ \frac{du}{dx} = u' &\rightarrow du = u' dx \end{aligned}$$

By rewriting this equation, you obtain the following theorem.  $uv = \int u \, dv + \int v \, du$

### THEOREM 8.1 INTEGRATION BY PARTS

If  $u$  and  $v$  are functions of  $x$  and have continuous derivatives, then

$$\int u \, dv = uv - \int v \, du.$$

This formula expresses the original integral in terms of another integral. Depending on the choices of  $u$  and  $dv$ , it may be easier to evaluate the second integral than the original one. Because the choices of  $u$  and  $dv$  are critical in the integration by parts process, the following guidelines are provided.

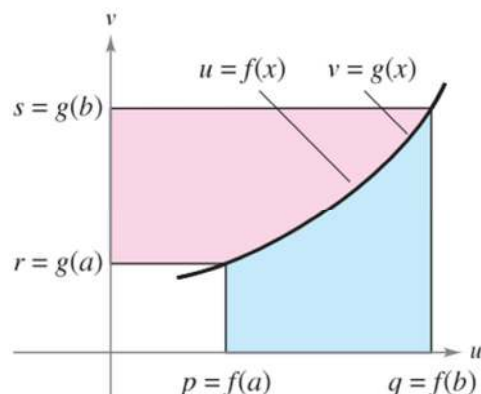
### GUIDELINES FOR INTEGRATION BY PARTS

1. Try letting  $dv$  be the most complicated portion of the integrand that fits a basic integration rule. Then  $u$  will be the remaining factor(s) of the integrand.
2. Try letting  $u$  be the portion of the integrand whose derivative is a function simpler than  $u$ . Then  $dv$  will be the remaining factor(s) of the integrand.

Note that  $dv$  always includes the  $dx$  of the original integrand.

$$\int u dv = uv - \int v du.$$

**Proof Without Words** Here is a different approach to proving the formula for integration by parts. Exercise taken from "Proof Without Words: Integration by Parts" by Roger B. Nelsen, *Mathematics Magazine*, 64, No. 2, April 1991, p. 130, by permission of the author.



Area  + Area

$$\int_r^s u dv + \int_p^q v du = \left[ uv \right]_{(p,r)}^{(q,s)}$$

$$\int_r^s u dv = \left[ uv \right]_{(p,r)}^{(q,s)} - \int_p^q v du$$

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## EXAMPLE 1 Integration by Parts

Find  $\int x e^x dx$ .

$$\int u dv = uv - \int v du.$$

**Solution** To apply integration by parts, you need to write the integral in the form  $\int u dv$ . There are several ways to do this.

$$\int \underbrace{(x)}_u \underbrace{(e^x dx)}_{dv}, \quad \int \underbrace{(e^x)}_u \underbrace{(x dx)}_{dv}, \quad \int \underbrace{(1)}_u \underbrace{(x e^x dx)}_{dv}, \quad \int \underbrace{(x e^x)}_u \underbrace{(dx)}_{dv}$$

the first option because the derivative of  $u = x$  is simpler than  $x$ , and  $dv = e^x dx$  is the most complicated portion of the integrand that fits a basic integration formula.

$$dv = e^x dx \Rightarrow v = \int dv = \int e^x dx = e^x$$

$$u = x \Rightarrow du = dx$$

Now, integration by parts produces

$$\int u dv = uv - \int v du$$

Integration by parts formula

$$\int x e^x dx = x e^x - \int e^x dx$$

Substitute.

$$= x e^x - e^x + C.$$

Integrate.

To check this, differentiate  $x e^x - e^x + C$  to see that you obtain the original integrand.

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## EXAMPLE 2 Integration by Parts

Find  $\int x^2 \ln x dx$ .

$$\int u dv = uv - \int v du.$$

**Solution** In this case,  $x^2$  is more easily integrated than  $\ln x$ . Furthermore, the derivative of  $\ln x$  is simpler than  $\ln x$ . So, you should let  $dv = x^2 dx$ .

$$dv = x^2 dx \Rightarrow v = \int x^2 dx = \frac{x^3}{3}$$

$$u = \ln x \Rightarrow du = \frac{1}{x} dx$$

Integration by parts produces

$$\int u dv = uv - \int v du$$

Integration by parts formula

$$\int x^2 \ln x dx = \frac{x^3}{3} \ln x - \int \left( \frac{x^3}{3} \right) \left( \frac{1}{x} \right) dx$$

Substitute.

$$= \frac{x^3}{3} \ln x - \frac{1}{3} \int x^2 dx$$

Simplify.

$$= \frac{x^3}{3} \ln x - \frac{x^3}{9} + C.$$

Integrate.

You can check this result by differentiating.

$$\frac{d}{dx} \left[ \frac{x^3}{3} \ln x - \frac{x^3}{9} \right] = \frac{x^3}{3} \left( \frac{1}{x} \right) + (\ln x)(x^2) - \frac{x^2}{3} = x^2 \ln x$$

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## EXAMPLE 4 Repeated Use of Integration by Parts

Find  $\int x^2 \sin x dx$ .

**Solution** The factors  $x^2$  and  $\sin x$  are equally easy to integrate. However, the derivative of  $x^2$  becomes simpler, whereas the derivative of  $\sin x$  does not. So, you should let  $u = x^2$ .

$$dv = \sin x dx \Rightarrow v = \int \sin x dx = -\cos x$$

$$u = x^2 \Rightarrow du = 2x dx$$

Now, integration by parts produces

$$\int x^2 \sin x dx = -x^2 \cos x + \int 2x \cos x dx.$$

First use of integration by parts

$$\int u dv = uv - \int v du.$$

This first use of integration by parts has succeeded in simplifying the original integral, but the integral on the right still doesn't fit a basic integration rule. To evaluate that integral, you can apply integration by parts again. This time, let  $u = 2x$ .

$$dv = \cos x dx \Rightarrow v = \int \cos x dx = \sin x$$

$$u = 2x \Rightarrow du = 2 dx$$

Now, integration by parts produces

Second use of integration by parts

$$\int 2x \cos x dx = 2x \sin x - \int 2 \sin x dx = 2x \sin x + 2 \cos x + C.$$

Combining these two results, you can write

$$\int x^2 \sin x dx = -x^2 \cos x + 2x \sin x + 2 \cos x + C.$$

not to interchange the substitutions in successive applications. the first substitution was  $u = x^2$  and  $dv = \sin x dx$ . the second application,  $u = \cos x$  and  $dv = 2x dx$ .

$$\int x^2 \sin x dx = -x^2 \cos x + \int 2x \cos x dx = -x^2 \cos x + x^2 \cos x + \int x^2 \sin x dx = \int x^2 \sin x dx$$

undoing the previous integration and returning to the original integral

watch for a constant multiple of the original integral.

As you gain experience in using integration by parts, your skill in determining  $u$  and  $dv$  will increase. The following summary lists several common integrals with suggestions for the choices of  $u$  and  $dv$ .

### SUMMARY OF COMMON INTEGRALS USING INTEGRATION BY PARTS

1. For integrals of the form

$$\int x^n e^{ax} dx, \quad \int x^n \sin ax dx, \quad \text{or} \quad \int x^n \cos ax dx$$

let  $u = x^n$  and let  $dv = e^{ax} dx$ ,  $\sin ax dx$ , or  $\cos ax dx$ .

2. For integrals of the form

$$\int x^n \ln x dx, \quad \int x^n \arcsin ax dx, \quad \text{or} \quad \int x^n \arctan ax dx$$

let  $u = \ln x$ ,  $\arcsin ax$ , or  $\arctan ax$  and let  $dv = x^n dx$ .

3. For integrals of the form

$$\int e^{ax} \sin bx dx \quad \text{or} \quad \int e^{ax} \cos bx dx$$

let  $u = \sin bx$  or  $\cos bx$  and let  $dv = e^{ax} dx$ .

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$$\begin{aligned} \frac{d}{dx} [\arcsin u] &= \frac{u'}{\sqrt{1-u^2}} \\ \frac{d}{dx} [\arccos u] &= \frac{-u'}{\sqrt{1-u^2}} \\ \frac{d}{dx} [\arctan u] &= \frac{u'}{1+u^2} \end{aligned}$$

**STUDY TIP** You can use the acronym **LIATE** as a guideline for choosing  $u$  in integration by parts. In order, check the integrand for the following.

Is there a Logarithmic part?

Is there an Inverse trigonometric part?

Is there an Algebraic part?

Is there a Trigonometric part?

Is there an Exponential part?

## Tabular Method

In problems involving repeated applications of integration by parts, a tabular method, illustrated in Example 7, can help to organize the work. This method works well for integrals of the form  $\int x^n \sin ax dx$ ,  $\int x^n \cos ax dx$ , and  $\int x^n e^{ax} dx$ .

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### EXAMPLE 7 Using the Tabular Method

Find  $\int x^2 \sin 4x dx$ .

**Solution** Begin as usual by letting  $u = x^2$  and  $dv = \sin 4x dx$ . Next, create a table consisting of three columns, as shown.

| Alternate Signs | $u$ and Its Derivatives | $v'$ and Its Antiderivatives |
|-----------------|-------------------------|------------------------------|
| +               | $x^2$                   | $\sin 4x$                    |
| -               | $2x$                    | $-\frac{1}{4} \cos 4x$       |
| +               | $2$                     | $-\frac{1}{16} \sin 4x$      |
| -               | $0$                     | $\frac{1}{64} \cos 4x$       |

Differentiate until you obtain 0 as a derivative.

The solution is obtained by adding the signed products of the diagonal entries:

$$\int x^2 \sin 4x dx = -\frac{1}{4} x^2 \cos 4x + \frac{1}{8} x \sin 4x + \frac{1}{32} \cos 4x + C.$$

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(1)

$$\int x^n e^{ax} dx, \quad \int x^n \sin ax dx, \quad \text{or} \quad \int x^n \cos ax dx \quad \textcircled{1}$$

### EXAMPLE

$$\int x \sin x dx \quad u = x \quad du = dx$$

$$dv = \sin(x) dx \quad v = -\cos(x)$$

$$\int u dv = uv - \int v du$$

$$= x(-\cos(x)) - \int -\cos(x) dx$$

$$= -x \cos(x) + \int \cos(x) dx$$

$$= -x \cos(x) + \sin(x) + C$$

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(3)

$$\int x^n e^{ax} dx, \quad \int x^n \sin ax dx, \quad \text{or} \quad \int x^n \cos ax dx \quad \textcircled{1}$$

**EXAMPLE**  $\int x^2 \cos x dx$   $u = x^2$   $du = 2x dx$   
 $dv = \cos x$   $v = \sin(x)$

$$\int u dv = uv - \int v du$$

$$\begin{aligned} &= x^2 \sin(x) - \int \sin(x) 2x dx \\ &= x^2 \sin(x) - 2 \int \sin(x) x dx \\ &= x^2 \sin(x) - 2(-x \cos(x) + \int \cos(x) dx) \\ &= x^2 \sin(x) - 2(-x \cos(x) + \sin(x)) \\ &= x^2 \sin(x) + 2x \cos(x) - 2 \sin(x) + C \end{aligned}$$

$u = x$   
 $dv = \sin(x)$   
 $du = dx$   
 $v = -\cos(x)$

(5)

$$\int x^n e^{ax} dx, \quad \int x^n \sin ax dx, \quad \text{or} \quad \int x^n \cos ax dx \quad \textcircled{1}$$

**EXAMPLE**  $\int x^4 e^x dx$  Using the Tabular Method  
 $u = x^4$   $dv = v' dx = e^x dx$   $v' = e^x$

|   |         |       |
|---|---------|-------|
| - | $x^4$   | $e^x$ |
| + | $4x^3$  | $e^x$ |
| - | $12x^2$ | $e^x$ |
| + | $24x$   | $e^x$ |
| - | $24$    | $e^x$ |
|   | $0$     | $e^x$ |

$$\int x^4 e^x dx = x^4 e^x - 4x^3 e^x + 12x^2 e^x - 24x e^x + C$$

(7)

$$\int x^n \ln x dx, \quad \int x^n \arcsin ax dx, \quad \text{or} \quad \int x^n \arctan ax dx \quad \textcircled{2}$$

**EXAMPLE**  $\int x^3 \ln x dx$   $u = \ln x$   $du = \frac{1}{x} dx$   
 $dv = x^3$   $v = \frac{x^4}{4}$

$$\begin{aligned} \frac{x^4}{4} \ln x - \int \frac{x^4}{4} \cdot \frac{1}{x} dx &= \frac{x^4 \ln x}{4} - \frac{1}{4} \int x^3 dx \\ &= \frac{x^4 \ln x}{4} - \frac{x^4}{16} + C \end{aligned}$$

(9)

$$\int e^{ax} \sin bx dx \quad \text{or} \quad \int e^{ax} \cos bx dx \quad \textcircled{3}$$

**EXAMPLE**  $\int e^x \cos x dx$   $u = \cos(x)$   $du = -\sin(x) dx$   
 $dv = e^x$   $v = e^x$

$$\begin{aligned} &e^x \cos(x) - \int e^x (-\sin(x)) dx \\ &= e^x \cos(x) + \int \sin(x) e^x dx \\ &= e^x \cos(x) + e^x \sin(x) - \int e^x \cos x dx \end{aligned}$$

$$\begin{aligned} 2 \int e^x \cos x dx &= e^x \cos(x) + e^x \sin(x) \\ \int e^x \cos x dx &= \frac{e^x \cos(x) + e^x \sin(x)}{2} \end{aligned}$$

$u = \sin x \Rightarrow du = \cos x dx$   
 $dv = e^x dx \Rightarrow v = e^x$   
 $e^x \sin(x) - \int e^x \cos x dx$

(9)

$$\int e^{ax} \sin bx \, dx \quad \text{or} \quad \int e^{ax} \cos bx \, dx \quad \textcircled{3}$$

**EXAMPLE**  $\int e^x \cos x \, dx$

by Substitution &amp; by part

**EXAMPLE**  $\int \cos(\sqrt{x}) \, dx$

$u = \sqrt{x} \Rightarrow \frac{du}{dx} = \frac{1}{2\sqrt{x}} \Rightarrow \frac{du}{dx} = \frac{1}{2u} \Rightarrow dx = 2u \, du$

$\int \cos(\sqrt{x}) \, dx = \int \cos(u) \cdot 2u \, du = 2 \int u \cos(u) \, du$

$\int x \cos(x) \, dx \xrightarrow{u=x, dv=\cos(x)} \int x \cos(x) \, dx \xrightarrow{v=\sin(x)} \int x \cos(x) \, dx = (x)(\sin(x)) - \int \sin(x) \, dx = x \sin(x) + \cos(x) + C$

by Substitution &amp; by part

**EXAMPLE**  $\int \cos(\sqrt{x}) \, dx$

Integration with respect to  $u$

$u = \sqrt{x} \rightarrow \frac{du}{dx} = \frac{1}{2} x^{-1/2} \rightarrow \frac{du}{dx} = \frac{1}{2\sqrt{x}} \rightarrow \frac{du}{dx} = \frac{1}{2u} \rightarrow dx = 2u \, du$

$\int \cos(\sqrt{x}) \, dx = \int \cos u (2u) \, du = 2 \int u (\cos u) \, du$  By substitution

By part

$u = x$

$\frac{du}{dx} = 1$

$dv = \cos x \, dx$

$v = \int dv = \int \cos x \, dx = \sin x$

$\int u \, dv = uv - \int v \, du$

$\int x \cos x \, dx = (x)(\sin x) - \int (\sin x) \, dx = x \sin x - (-\cos x) + C = x \sin x + \cos x + C$

by Substitution &amp; by part

**EXAMPLE**  $\int \cos(\sqrt{x}) \, dx$

Integration with respect to  $u$

$u = \sqrt{x} \rightarrow \frac{du}{dx} = \frac{1}{2} x^{-1/2} \rightarrow \frac{du}{dx} = \frac{1}{2\sqrt{x}} \rightarrow \frac{du}{dx} = \frac{1}{2u} \rightarrow dx = 2u \, du$

$\int \cos(\sqrt{x}) \, dx = \int \cos u (2u) \, du = 2 \int u (\cos u) \, du$  By substitution

By part

$= 2(u \sin u + \cos u) + C$

$= 2(\sqrt{x} \sin \sqrt{x} + \cos \sqrt{x}) + C$

Math  $\int x \cos x \, dx = x \sin x + \cos x + C$



# BREAK

## APPLICATIONS OF INTEGRATION

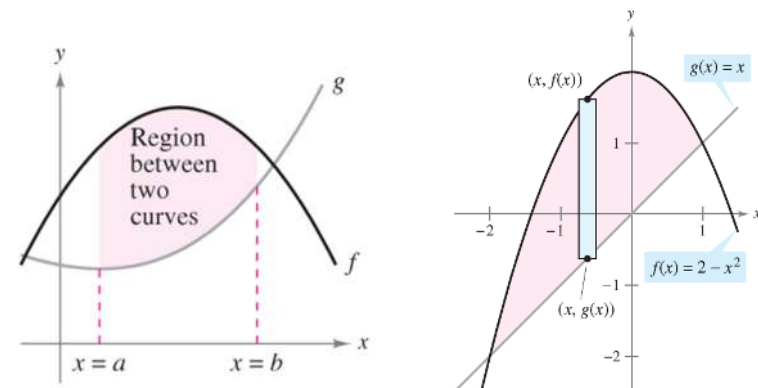
### Applications of Integration (I)

#### Objectives

- Use a definite integral to find the area of a region bounded by two curves.
- Find the volume of a solid of revolution.

### Area of a Region Between Two Curves

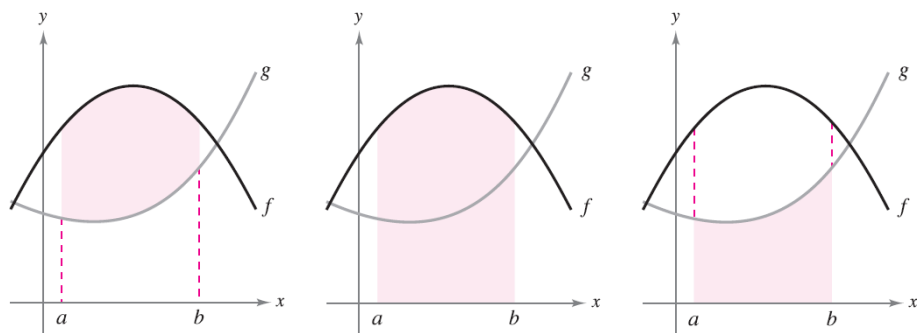
- Find the area of a region between two curves using integration.
- Find the area of a region between intersecting curves using integration.



## Area of a Region Between Two Curves

Previously ...

area of a region *under* a curve  $\rightarrow$  area of a region *between* two curves



Area of region  
between  $f$  and  $g$

Area of region  
under  $f$

Area of region  
under  $g$

$$\int_a^b [f(x) - g(x)] dx$$

$$= \int_a^b f(x) dx$$

$$- \int_a^b g(x) dx$$

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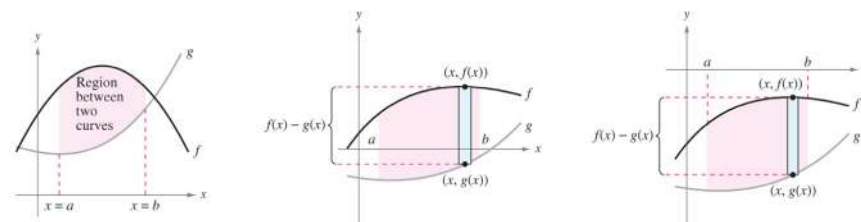
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## Area of a Region Between Two Curves

### AREA OF A REGION BETWEEN TWO CURVES

If  $f$  and  $g$  are continuous on  $[a, b]$  and  $g(x) \leq f(x)$  for all  $x$  in  $[a, b]$ , then the area of the region bounded by the graphs of  $f$  and  $g$  and the vertical lines  $x = a$  and  $x = b$  is

$$A = \int_a^b [f(x) - g(x)] dx.$$



continuous and  $g(x) \leq f(x)$  for all  $x$  in the interval  $[a, b]$

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### EXAMPLE 1 Finding the Area of a Region Between Two Curves

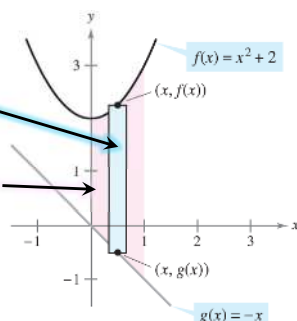
Find the area of the region bounded by the graphs of  $y = x^2 + 2$ ,  $y = -x$ ,  $x = 0$ , and  $x = 1$ .

**Solution** Let  $g(x) = -x$  and  $f(x) = x^2 + 2$ . Then  $g(x) \leq f(x)$  for all  $x$  in  $[0, 1]$ . So, the area of the representative rectangle is

$$\begin{aligned} \Delta A &= [f(x) - g(x)] \Delta x \\ &= [(x^2 + 2) - (-x)] \Delta x \end{aligned}$$

and the area of the region is

$$\begin{aligned} A &= \int_a^b [f(x) - g(x)] dx = \int_0^1 [(x^2 + 2) - (-x)] dx \\ &= \left[ \frac{x^3}{3} + \frac{x^2}{2} + 2x \right]_0^1 \\ &= \frac{1}{3} + \frac{1}{2} + 2 \\ &= \frac{17}{6}. \end{aligned}$$



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### EXAMPLE 2 A Region Lying Between Two Intersecting Graphs

Find the area of the region bounded by the graphs of  $f(x) = 2 - x^2$  and  $g(x) = x$ .

**Solution** notice that the graphs of  $f$  and  $g$  have two points of intersection. To find the  $x$ -coordinates of these points, set  $f(x)$  and  $g(x)$  equal to each other and solve for  $x$ .

$$\begin{aligned} 2 - x^2 &= x \\ -x^2 - x + 2 &= 0 \\ -(x + 2)(x - 1) &= 0 \\ x &= -2 \text{ or } 1 \end{aligned}$$

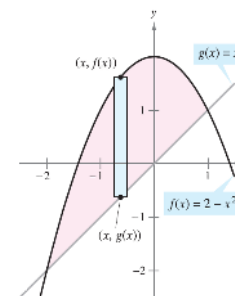
Set  $f(x)$  equal to  $g(x)$ .  
Write in general form.  
Factor.  
Solve for  $x$ .  
✓ ๔๔๓๓๓

So,  $a = -2$  and  $b = 1$ . Because  $g(x) \leq f(x)$  for all  $x$  in the interval  $[-2, 1]$ , the representative rectangle has an area of

$$\begin{aligned} \Delta A &= [f(x) - g(x)] \Delta x \\ &= [(2 - x^2) - x] \Delta x \end{aligned}$$

and the area of the region is

$$\begin{aligned} A &= \int_{-2}^1 [(2 - x^2) - x] dx = \left[ -\frac{x^3}{3} - \frac{x^2}{2} + 2x \right]_{-2}^1 \\ &= \frac{9}{2}. \end{aligned}$$



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### EXAMPLE 4 Curves That Intersect at More Than Two Points

Find the area of the region between the graphs of  $f(x) = 3x^3 - x^2 - 10x$  and  $g(x) = -x^2 + 2x$ .

?

$$3x^3 - x^2 - 10x = -x^2 + 2x$$

$$3x^3 - 12x = 0$$

$$3x(x^2 - 4) = 0$$

$$3x(x-2)(x+2) = 0$$

$$x = -2, 0, 2$$

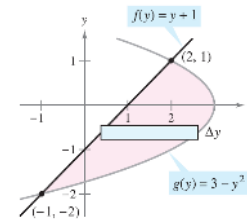
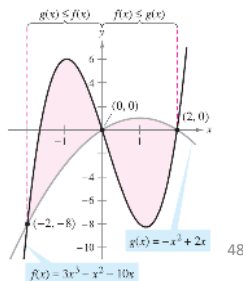
$$g(x) \leq f(x) \rightarrow [-2, 0]$$

$$f(x) \leq g(x) \rightarrow [0, 2]$$

$$A = \int_{-2}^0 [f(x) - g(x)] dx + \int_0^2 [g(x) - f(x)] dx$$

$$= \int_{-2}^0 (3x^3 - 12x) dx + \int_0^2 (-3x^3 + 12x) dx$$

$$= \left[ \frac{3x^4}{4} - 6x^2 \right]_{-2}^0 + \left[ -\frac{3x^4}{4} + 6x^2 \right]_0^2 = 24$$



If the graph of a function of  $y$  is a boundary of a region, it is often convenient to use representative rectangles that are horizontal and find the area by integrating with respect to  $y$ . In general, to determine the area between two curves, you can use

$$A = \int_{x_1}^{x_2} [(\text{top curve}) - (\text{bottom curve})] dx$$

in variable  $x$

Vertical rectangles

$$A = \int_{y_1}^{y_2} [(\text{right curve}) - (\text{left curve})] dy$$

in variable  $y$

Horizontal rectangles

where  $(x_1, y_1)$  and  $(x_2, y_2)$  are either adjacent points of intersection of the two curves involved or points on the specified boundary lines.

### EXAMPLE 5 Horizontal Representative Rectangles

Find the area of the region bounded by the graphs of  $x = 3 - y^2$  and  $x = y + 1$ .

**Solution** Consider

$$g(y) = 3 - y^2 \quad \text{and} \quad f(y) = y + 1.$$

a function of  $y$

These two curves intersect when  $y = -2$  and  $y = 1$ , as shown in Figure  $f(y) \leq g(y)$  on this interval, you have

$$\Delta A = [g(y) - f(y)] \Delta y = [(3 - y^2) - (y + 1)] \Delta y.$$

So, the area is

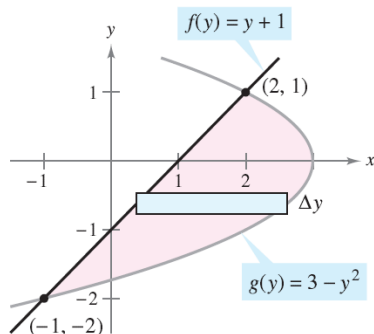
$$A = \int_{-2}^1 [(3 - y^2) - (y + 1)] dy$$

$$= \int_{-2}^1 (-y^2 - y + 2) dy$$

$$= \left[ -\frac{y^3}{3} - \frac{y^2}{2} + 2y \right]_{-2}^1$$

$$= \left( -\frac{1}{3} - \frac{1}{2} + 2 \right) - \left( \frac{8}{3} - 2 - 4 \right)$$

$$= \frac{9}{2}.$$



notice that by integrating with respect to  $y$  you need only one integral. If you had integrated with respect to  $x$ , you would have needed two integrals because the upper boundary would have changed at  $x = 2$ , as shown in Figure

$$A = \int_{-1}^2 (x - 1) - (-\sqrt{3-x}) dx + \int_2^3 (\sqrt{3-x}) - (-\sqrt{3-x}) dx$$

$$= \int_{-1}^2 [(x - 1) + \sqrt{3-x}] dx + \int_2^3 (\sqrt{3-x} + \sqrt{3-x}) dx$$

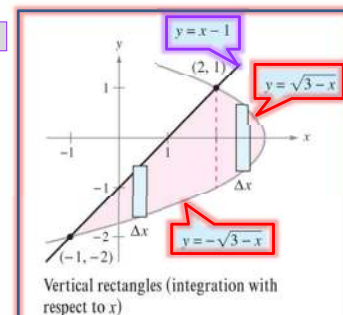
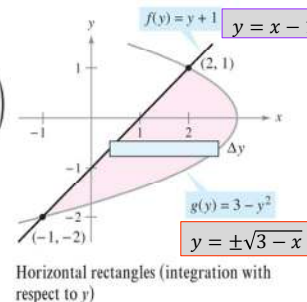
$$= \int_{-1}^2 [x - 1 + (3-x)^{1/2}] dx + 2 \int_2^3 (3-x)^{1/2} dx$$

$$= \left[ \frac{x^2}{2} - x - \frac{(3-x)^{3/2}}{3/2} \right]_{-1}^2 - 2 \left[ \frac{(3-x)^{3/2}}{3/2} \right]_2^3$$

$$= \left( 2 - 2 - \frac{2}{3} \right) - \left( \frac{1}{2} + 1 - \frac{16}{3} \right) - 2(0) + 2\left( \frac{2}{3} \right)$$

$$= \frac{9}{2}$$

More difficult



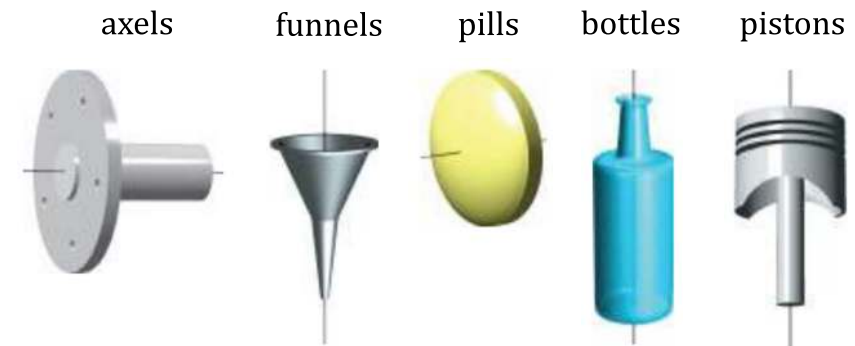


# Volume: The Disk Method

- Find the volume of a solid of revolution using the disk method.
- Find the volume of a solid of revolution using the washer method.

# The Disk Method

Previously ...  
area  $\rightarrow$  volume    3-dimensional solid whose cross sections are similar



Solids of revolution

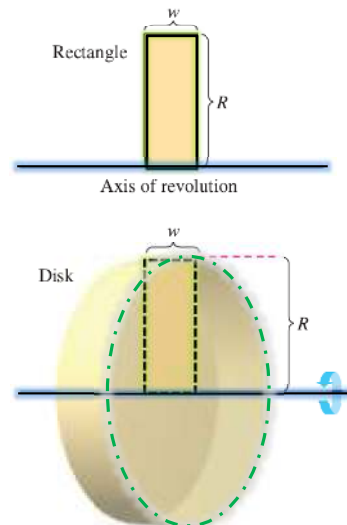
A solid of revolution

axis of revolution

# The Disk Method

The simplest such solid is a right circular cylinder or disk.  
It is formed by revolving a **rectangle** about an **axis** adjacent to one side of the rectangle.

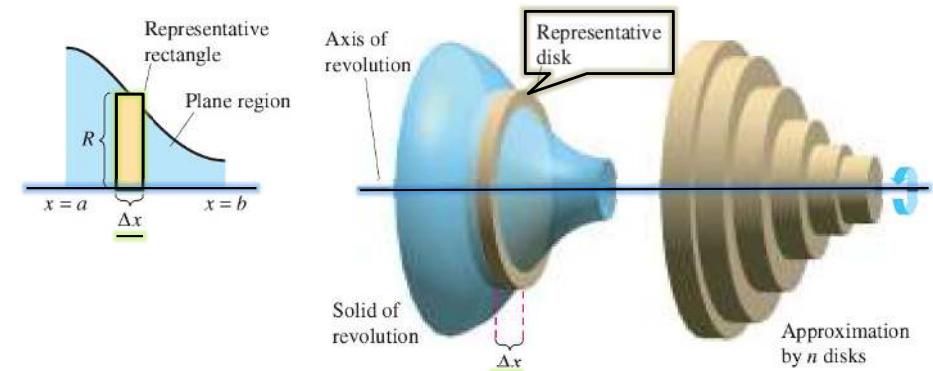
$$\begin{aligned}\text{Volume of disk} &= (\text{area of disk})(\text{width of disk}) \\ &= \pi R^2 w\end{aligned}$$



# The Disk Method

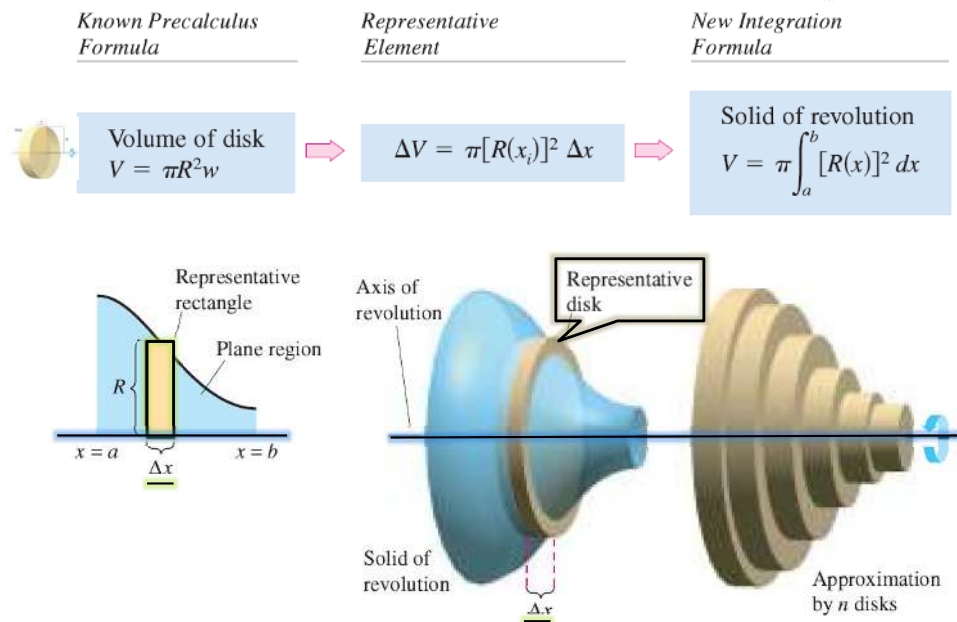
To find the volume of a **general solid of revolution** ... (see figure)  
revolve a representative **rectangle** about the **axis** of revolution,  
it generates a **representative disk** whose **volume** is ...

$$\Delta V = \pi R^2 \Delta x.$$



Approximation by  $n$  disks

# The Disk Method



## THE DISK METHOD

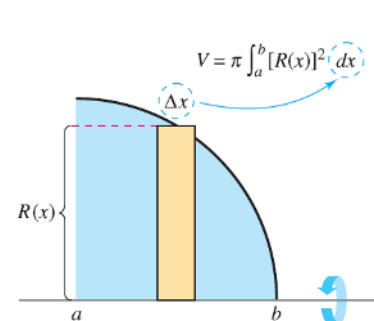
To find the volume of a solid of revolution with the **disk method**, use one of the following,

Horizontal Axis of Revolution

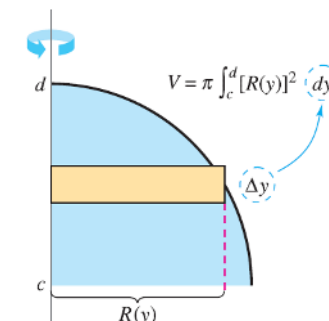
$$\text{Volume} = V = \pi \int_a^b [R(x)]^2 dx$$

Vertical Axis of Revolution

$$\text{Volume} = V = \pi \int_c^d [R(y)]^2 dy$$



Horizontal axis of revolution



Vertical axis of revolution

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### EXAMPLE 1 Using the Disk Method

Find the volume of the solid formed by revolving the region bounded by the graph of

$$f(x) = \sqrt{\sin x}$$

and the  $x$ -axis ( $0 \leq x \leq \pi$ ) about the  $x$ -axis.

**Solution** From the representative rectangle in the upper graph the radius of this solid is

$$\begin{aligned} R(x) &= f(x) \\ &= \sqrt{\sin x}. \end{aligned}$$

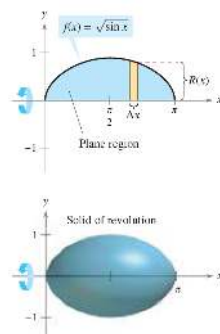
So, the volume of the solid of revolution is

$$\begin{aligned} V &= \pi \int_a^b [R(x)]^2 dx = \pi \int_0^\pi (\sqrt{\sin x})^2 dx \\ &= \pi \int_0^\pi \sin x dx \\ &= \pi [-\cos x]_0^\pi \\ &= \pi(1 + 1) \\ &= 2\pi. \end{aligned}$$

Apply disk method.

Simplify

Integrate.



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### EXAMPLE 2 Revolving About a Line That Is Not a Coordinate Axis

Find the volume of the solid formed by revolving the region bounded by

$$f(x) = 2 - x^2$$

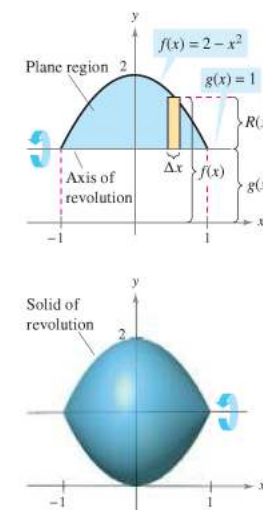
and  $g(x) = 1$  about the line  $y = 1$ , as shown in Figure

$$R(x) = f(x) - g(x)$$

$$= 2 - x^2 - 1$$

$$= 1 - x^2$$

$$\begin{aligned} V &= \pi \int_a^b [R(x)]^2 dx = \pi \int_{-1}^1 (1 - x^2)^2 dx \\ &= \pi \int_{-1}^1 (1 - 2x^2 + x^4) dx \\ &= \pi \left[ x - \frac{2x^3}{3} + \frac{x^5}{5} \right]_{-1}^1 \\ &= \frac{16\pi}{15} \end{aligned}$$



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# The Washer Method

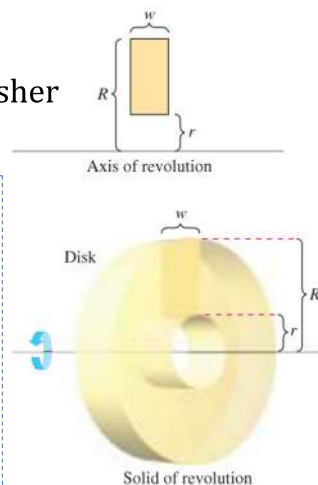
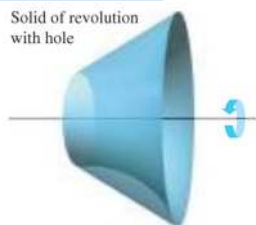
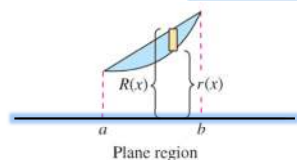
Solids of revolution with hole

Representative disk → Representative washer

$$\text{Volume of washer} = \pi(R^2 - r^2)w$$

If a region bounded by an **outer radius**  $R(x)$  and an **inner radius**  $r(x)$ , and the region is revolved about its axis of revolution, the volume of the resulting solid is given by

$$V = \pi \int_a^b ([R(x)]^2 - [r(x)]^2) dx.$$



the volume of the outer radius  
—  
the volume of the inner radius

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## EXAMPLE 3 Using the Washer Method

Find the volume of the solid formed by revolving the region bounded by the graphs of  $y = \sqrt{x}$  and  $y = x^2$  about the  $x$ -axis, as shown in Figure

**Solution** the outer and inner radii are as follows.

$$R(x) = \sqrt{x}$$

$$r(x) = x^2$$

Integrating between 0 and 1 produces

$$\begin{aligned} V &= \pi \int_a^b ([R(x)]^2 - [r(x)]^2) dx \\ &= \pi \int_0^1 [(\sqrt{x})^2 - (x^2)^2] dx \\ &= \pi \int_0^1 (x - x^4) dx \\ &= \pi \left[ \frac{x^2}{2} - \frac{x^5}{5} \right]_0^1 \\ &= \frac{3\pi}{10} \end{aligned}$$

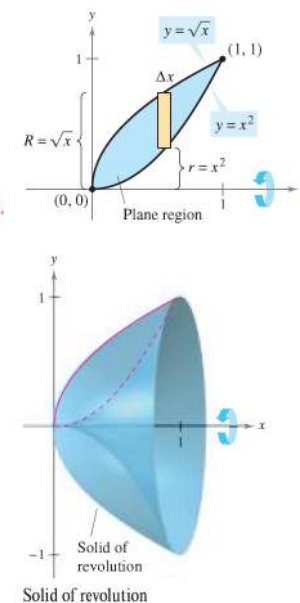
Outer radius

Inner radius

Apply washer method.

Simplify.

Integrate.



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## Homework

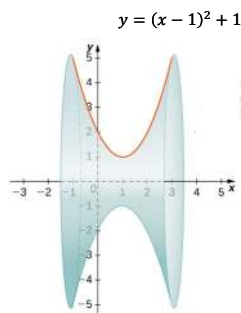
1) Find

a)  $\int x^3 \cos x \, dx$

b)  $\int x^5 \ln x \, dx$

c)  $\int e^x \sin x \, dx$

2) Find the volume of the solid generated by revolving the region bounded by  $y = (x-1)^2 + 1$  and the  $x$ -axis on  $[-1, 3]$  about the  $x$ -axis. (using the **disk method**)



- Write down a step-by-step solution in a piece of A4 paper (or on an electronic paper).
- **DO NOT TYPE.**
- Submit your solution in MS Team.
- Do not forget to write your name and student ID.